

JSS2 Mathematics Diagnostic Assessment (Part 3)

Curriculum: Cambridge Lower Secondary (Stages 7 & 8) & Nigerian JSS2
Cluster 3: Geometry, Mensuration & Data Handling (Questions 41 - 60)

Mathematics - Cluster 3

No.	Question & Options	Ans	Diagnostic Focus / Concept Tag
41	The sum of angles on a straight line is: A) 90° B) 180° C) 270° D) 360°	B	Basic angle properties.
42	Find the value of x in a triangle with angles 40° , 60° , and x . A) 80° B) 100° C) 180° D) 90°	A	Sum of angles in a triangle.
43	Calculate the area of a circle with a radius of 7cm (Take $\pi = \frac{22}{7}$). A) 44cm^2 B) 154cm^2 C) 49cm^2 D) 14cm^2	B	Mensuration (Circle area).
44	In a right-angled triangle, if the legs are 3cm and 4cm, find the hypotenuse. A) 5cm B) 7cm C) 12cm D) 25cm	A	Pythagoras' Theorem (Cambridge Stage 8).
45	Which of these shapes has exactly 4 lines of symmetry? A) Rectangle B) Square C) Circle D) Kite	B	Symmetry and properties of 2D shapes.
46	Find the exterior angle of a regular hexagon. A) 60° B) 120° C) 45° D) 90°	A	Polygons (Exterior angles).
47	Calculate the volume of a cuboid with dimensions $10\text{cm} \times 5\text{cm} \times 2\text{cm}$. A) 100cm^3 B) 17cm^3 C) 50cm^3 D) 70cm^3	A	3D Geometry (Volume).
48	The bearing of point B from A is 045° . What is the back bearing? A) 135° B) 225° C) 315° D) 045°	B	Bearings and direction (Nigerian Curriculum).
49	Find the mean of the numbers: 4, 7, 10, 3, 6. A) 6 B) 7 C) 5 D) 30	A	Statistics (Arithmetic Mean).
50	What is the mode of the set: {2, 5, 2, 8, 9, 2, 5}? A) 2 B) 5 C) 8 D) 9	A	Statistics (Mode).
51	Calculate the perimeter of a semi-circle with diameter 14cm (Take $\pi = \frac{22}{7}$). A) 36cm B) 22cm C) 44cm D) 58cm	A	Mensuration (Compound perimeter).
52	Two angles are supplementary. If one is 75° , find the other. A) 15° B) 105° C) 285° D) 180°	B	Angle relationships.
53	Find the surface area of a cube with side length 3cm. A) 27cm^2 B) 54cm^2 C) 9cm^2 D) 36cm^2	B	3D Geometry (Surface Area).
54	If the probability of it raining is 0.3, what is the probability it does not rain? A) 0.7 B) 0 B) 1.3 D) 0.3	A	Probability basics.
55	A bag contains 3 red balls and 7 blue balls. A ball is picked at random. Find the probability it is red.	C	Probability of simple events.

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	A) $\frac{3}{7}$ B) $\frac{10}{3}$ C) $\frac{3}{10}$ D) $\frac{7}{10}$		
56	Find the median of: 12, 15, 10, 18, 11. A) 10 B) 12 C) 15 D) 18	B	Statistics (Median).
57	An isosceles triangle has one angle of 100° . If this is the apex angle, find the base angles. A) 40° each B) 80° each C) 100° each D) 50° each	A	Triangle properties.
58	Convert 2.5 Liters to Milliliters. A) 250ml B) 2500ml C) 25ml D) 0.0025ml	B	Standard units conversion.
59	How many edges does a triangular prism have? A) 6 B) 9 C) 5 D) 12	B	Visualizing 3D shapes.
60	Diagnostic Insight: Success in Geometry and Pythagoras (Q44) is essential for Technical Drawing and Physics.	-	Pathway alignment.